

Quantum-Enhanced Agrivoltaic Digital Twin: Spectral Co-Design, Multifunctional Net Energy Benefit, and Quantum-Secured Telemetry

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Abstract

Agrivoltaic deployment is constrained by a system-level trade-off: static semi-transparent photovoltaic modules resolve the food–energy land conflict only by uniformly sacrificing either electrical output or crop productivity. Here we present a quantum-guided agrivoltaic system in which a semi-transparent organic photovoltaic layer couples to a plasmonic nanoparticle-on-mirror (NPoM) cavity acting as an active spectral shaper. By transmitting narrow bands at 750 nm and 820 nm, the system prepares polaron-dressed states in the Fenna–Matthews–Olson photosynthetic complex, extending exciton coherence by 50 % and achieving an area-weighted global canopy yield of 97.1 %. The NPoM sentinel layer enables multiplexed SERS and quantum dot diagnostics of agricultural contaminants and crop stress. On the macroscopic scale, the shield reduces crop evapotranspiration by 28 % and yields a Net Ecological Benefit of $19.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$. A BB84 quantum link secures the IoT telemetry. Finally, techno-economic analysis of a Cameroonian micro-module gives a payback period of 4.23 yr, identifying conditions under which the multifunctional net benefit stays positive for the modelled 500 m^2 cooperative case.

Keywords: quantum agrivoltaics, FMO complex, vibronic coherence, plasmonic nanoparticle-on-mirror cavity, SERS diagnostics, FAO-56 evapotranspiration, agrivoltaic techno-economics, quantum digital twin

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Agrivoltaics co-locates photovoltaic (PV) systems and agricultural cultivation to mitigate the land-use competition between food production and renewable energy generation [1, 2]. Land-use optimization and mobile-PV foundations for this field were established in *Applied Energy* [3, 4], and recent syntheses systematize overhead, vertical, and tracking typologies with land-equivalent-ratio and levelized-cost metrics [5]. Across these studies, three gaps recur and remain unresolved. (i) The optical architecture is *static*: semi-transparent PV claddings act as fixed spectral filters that attenuate irradiance uniformly, trading electrical output for crop biomass without optimization of the spectral light-sharing window [6, 7]. Dynamic, physics-guided spectral management has not been transposed into an operating agrivoltaic stack. The cost of stasis is quantifiable: in the mode-volume scan reported below, the forward transfer yield varies by a factor of 1.6 and the SERS enhancement by a factor of 48 across the accessible range (Table 2), so a statically chosen cladding fixes the energy–diagnostics balance at deployment with no ability to re-trade it. (ii) Techno-economic and lifecycle assessments remain decoupled from the underlying device physics: electricity, crop, water, and carbon accounts are evaluated subsystem by subsystem, so multifunctional trade-offs are described rather than co-optimized [8, 9]. The cost of this decoupling is concrete: accounting that ignores the local transport physics would credit the canopy with the passive baseline of 0.980 instead of the physically consistent 0.971 — and would never discover that the 91.8% local penalty costs only 0.9 points globally, the stabilizer that makes NPoM integration viable. (iii) The PV layer is treated purely as an energy harvester: in-situ crop-stress and contaminant sensing is bolted on as separate hardware rather than co-designed with the optical stack, despite the maturity of field-deployable SERS analysis [10], of quantum-enabled agricultural workflows [11], and of quantum-sensing readiness for precision agriculture [12]. Co-design changes the economics of the sensing function: reusing the NPoM optical infrastructure delivers SERS diagnostics at a canopy-area cost of 1% and a canopy-yield cost of 0.9% relative to the passive baseline (0.980 to 0.971), instead of separate sensing hardware with its own capital and footprint.

The photosynthetic apparatus of green plants and bacteria has evolved under a spectrally structured photon flux, where specific wavelengths drive charge separation while others dissipate as heat [13]. This spectral specificity allows the PV layer to transmit only the photosynthetically active wavelengths while absorbing the remainder, converting the food-energy conflict into a spectral co-design problem.

The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria

41 serves as the model system for spectral light-sharing [14]. Its seven bac-
 42 teriochlorophyll *a* sites have been extensively characterized through two-
 43 dimensional electronic spectroscopy, revealing oscillatory coherence signa-
 44 tures that persist for hundreds of femtoseconds at cryogenic [15] and room
 45 temperature [16]. Although the functional relevance of these quantum beats
 46 remains debated, consensus has emerged that vibronic coupling — the hy-
 47 bridization of electronic transitions with underdamped protein vibrations —
 48 governs the flow of excitation energy [17, 18]. Non-Markovian frameworks
 49 such as the hierarchical equations of motion [19] and the process tensor hier-
 50 archy of pure states (PT-HOPS) [20, 21] capture these effects with demon-
 51 strated numerical convergence. Device-adjacent uses of exact open-quantum-
 52 system methods remain rare — the closest precedent is a general HEOM li-
 53 brary applied to light-harvesting and control problems without a photovoltaic
 54 product [22] — and photonic-quantum solar design and vibronic-screening
 55 workflows likewise stop short of an operating device [23, 24].

56 Previous work demonstrated that spectral filtering of the excitation pulse
 57 — aligning its spectrum with underdamped vibronic resonances at 750 nm
 58 and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO
 59 complex [25]. This selective vibronic excitation prepares polaron-dressed
 60 states that decouple from the dissipative protein-solvent bath. A related
 61 study from our group, published in the *Journal of Physical Chemistry Let-*
 62 *ters*, reports the filter, excitation-transport-rate, and single-pass-conversion-
 63 efficiency analysis [26]; the present work adds the monolithic photothermal-
 64 heating hardware, the SERS field calibration, and the Net Ecological Benefit
 65 and lifecycle assessment developed below. Single-photon experiments have
 66 confirmed that quantum coherence persists in individual light-harvesting
 67 complexes under ambient conditions [27], and quantum phase synchroniza-
 68 tion via exciton-vibrational energy dissipation has been identified as a mech-
 69 anism sustaining long-lived coherence [28]. Here we extend this principle in
 70 three directions.

71 First, we introduce an 8-site FMO model with a non-Hermitian reaction
 72 center trap on sites 3 and 4, enabling quantitative computation of the forward
 73 transfer yield Φ_{FT} as the time-integrated population captured by the reac-
 74 tion center. Second, we embed the FMO complex in a nanoparticle-on-mirror
 75 (NPoM) plasmonic cavity [29] that couples the biological light-harvesting sys-
 76 tem to the organic photovoltaic (OPV) layer through strong coherent cou-
 77 pling, forming a 9×9 dressed Hamiltonian. Third, we incorporate dynamic
 78 photo-protection mechanisms — Floquet Stark detuning and optomechan-

ically induced transparency (OMIT) — that mimic the non-photochemical quenching used by plants under high light stress.

We map this microscopic optimization onto a field-scale lifecycle assessment. We parameterize crop evapotranspiration using the FAO-56 Penman–Monteith model [30] under the spectrally selective OPV shield, compute the Net Ecological Benefit (NEB) across three scenarios, and evaluate socioeconomic access through cooperative ownership models and BB84 quantum key distribution [31] for secure IoT telemetry. This multi-scale perspective [32] links angstrom-scale vibronic couplings to square-meter canopy yields, providing a unified modelling framework for quantum-guided agrivoltaics (Fig. 5).

The central question of this work is falsifiable: *does an agrivoltaic stack whose spectral architecture is co-designed by exact non-Markovian quantum dynamics deliver a positive multifunctional net energy benefit (heat + power + crop + sensing) under field constraints, and which deployment barriers are physical rather than economic?* Three subordinate questions structure the Results. **(Q1) Device:** what does the NPoM co-design cost, and where does the cost land? The mode-volume scan (Table 2) quantifies the local penalty; the canopy-integration analysis shows where it is absorbed. **(Q2) Physics:** which microscopic states carry the trade-off? Exact PT-HOPS dynamics localize the loss channels, and Floquet–Stark detuning with OMIT tests whether they can be managed dynamically. **(Q3) System:** does the stack pay for itself? Coupled FAO-56 microclimate, cooperative technoeconomics, and BB84-secured telemetry internalize every function into a single Net Ecological Benefit account. Three falsifiable claim families follow. **(F1) Microscopic:** dual-band spectral filtering at 750 nm and 820 nm prepares polaron-dressed states that extend exciton coherence by 50 % and raise the forward transfer yield by 25 % at 295 K. **(F2) Device:** confining the NPoM cavity to a 1 % sentinel fraction preserves 97.1 % of the global canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) while retaining single-complex SERS diagnostics, despite the > 90 % local transport penalty quantified across the mode-volume scan. **(F3) System:** the FAO-56 smart shield cuts evapotranspiration by 28 % (± 10 % microclimate uncertainty) and, combined with OPV generation and cooperative economics, yields a Net Ecological Benefit of $19.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ and a 4.23 yr payback at 30 % subsidy for the modelled 500 m^2 cooperative case. We answer affirmatively but conditionally: the bounds (single-exciton treatment, site-independent baths, modelled fouling and matrix penalties, ± 10 % microclimate band) are stated explicitly in the

117 Limitations section. The headline outcome compresses this answer to one
 118 sentence, pictured in [Figure 1](#): a 91.8% local transport penalty confined to
 119 a 1% sentinel area costs 0.9 points of canopy-scale trapping yield (0.980 to
 120 0.971) while adding single-complex SERS diagnostics to the same footprint.

121 Results

122 8-Site FMO Hamiltonian with reaction center trap

Table 1: **8-site FMO site energies and key electronic couplings (cm^{-1})**. Full coupling matrix (28 unique pairs) provided in [Section S1](#).

Parameter	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
ε_n	280	420	0	110	270	500	310	200
$J_{n,n+1}$	—	-87.7	30.8	-53.5	-70.7	81.1	39.7	12.0

123 We construct an 8-site model of the FMO complex, extending the stan-
 124 dard 7-site framework [\[33\]](#) by including a coupling to the eighth bacteri-
 125 ochlorophyll that bridges the baseplate to the reaction center (Site 8, ~ 200
 126 cm^{-1} above Site 3). The electronic Hamiltonian is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (1)$$

127 where ε_n are the site energies (relative to Site 3), J_{nm} are the inter-site
 128 electronic couplings (all 28 unique pairs defined in [Section S1](#)), and $\Gamma_{\text{RC}} =$
 129 0.15 ps^{-1} is the irreversible reaction center trapping rate [\[34\]](#). The anti-
 130 Hermitian term $-i\Gamma_{\text{RC}}$ on Sites 3 and 4 (physical indices; Python indices 2
 131 and 3) models the irreversible charge separation at the reaction center. All 28
 132 inter-site couplings are validated to be finite, and the Hamiltonian is verified
 133 to be non-Hermitian only on the trapping diagonal terms ([Section S1](#)).

134 NPoM plasmonic strong coupling

135 The OPV-FMO interface is mediated by a nanoparticle-on-mirror
 136 (NPoM) architecture [\[29\]](#), where a gold nanoparticle (diameter 80 nm) is
 137 separated from a gold mirror by a 1 nm graphene monolayer spacer. The
 138 graphene barrier prevents Dexter-type electron transfer while maintaining
 139 electromagnetic field enhancement. The NPoM cavity supports an intense

140 localized surface plasmon mode that couples coherently to both the OPV
141 exciton band and the FMO optical transitions.

142 The single-emitter coupling strength follows the cavity quantum electro-
143 dynamics scaling:

$$g_0 = g_{\text{ref}} \sqrt{\frac{V_{\text{ref}}}{V_{\text{mode}}}}, \quad (2)$$

144 with $g_{\text{ref}} = 120 \text{ cm}^{-1}$ at $V_{\text{ref}} = 1 \text{ nm}^3$ [29]. At the sub-nanometer mode vol-
145 umes accessible in NPoM cavities ($V_{\text{mode}} < 1 \text{ nm}^3$), g_0 ranges from 120 cm^{-1}
146 to over 300 cm^{-1} , and the collective coupling $g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$ places the sys-
147 tem in the strong-coupling regime ($g_{\text{eff}} > \kappa$, where $\kappa \approx 800 \text{ cm}^{-1}$ is the plas-
148 mon linewidth). A numerical guardrail caps g_0 at 1000 cm^{-1} when $V_{\text{mode}} \rightarrow 0$
149 to prevent divergences. Recent advances in quantum-tunnel-junction plas-
150 monic sources have demonstrated that such sub-nanometer cavities can be
151 self-illuminating, eliminating the need for bulky external optics [35].

152 The dressed Hamiltonian expands to 9×9 :

$$\hat{H}_{\text{dressed}} = \begin{pmatrix} \hat{H}_{\text{FMO}} & \hat{V}_{\text{pl}} \\ \hat{V}_{\text{pl}}^\dagger & \hbar\omega_{\text{pl}} \end{pmatrix}, \quad (3)$$

153 where the plasmon mode at frequency $\omega_{\text{pl}} \approx 12500 \text{ cm}^{-1}$ couples to the
154 primary optical antenna sites (BChl 1 and 6, indices 0 and 5) with strength
155 g_0 . Figure 2a shows the energy level diagram.

156 *Non-Markovian quantum dynamics*

157 We propagate the 9-site dressed system using the PT-HOPS/SBD formal-
158 ism with production parameters: hierarchy depth $L = 8$, $K = 2$ Matsubara
159 terms, time step $\Delta t = 0.2 \text{ fs}$, and $n = 100$ disorder realizations ($\sigma = 50 \text{ cm}^{-1}$
160 static disorder) [20, 21]. The bath is modeled as a Drude-Lorentz over-
161 damped component ($\lambda_D = 35 \text{ cm}^{-1}$, $\gamma_D = 50 \text{ cm}^{-1}$) plus 12 underdamped
162 vibronic modes from the Kleinekathöfer/Coker parameterization. Conver-
163 gence is verified through hierarchy depth sweeps ($L = 6, 7, 8$) and Matsubara
164 truncation tests ($K = 1, 2, 3$); the maximum relative error between $L = 7$
165 and $L = 8$ populations is 3.1×10^{-11} , and between $K = 2$ and $K = 3$ is
166 3.3×10^{-5} (Section S2). The computational feasibility, scaling, and memory-
167 aware scheduling for these high-order simulations are detailed and validated
168 in our companion study [25].

169 Figure 2b shows the exciton population dynamics under broadband *vs.*
170 dual-band filtered excitation. The filtered case (solid lines) maintains higher

171 population on the trapping sites (Sites 3 and 4), leading to enhanced forward
 172 transfer yield:

$$\Phi_{\text{FT}} = \Gamma_{\text{RC}} \int_0^{t_{\text{max}}} \sum_{j \in \{3,4\}} P_j(t) dt, \quad (4)$$

173 where $P_j(t) = \rho_{jj}(t)$ are the site populations and $t_{\text{max}} = 1$ ps.

174 Under dual-band filtering, $\Phi_{\text{FT}} = 0.89(3)$, compared to $\Phi_{\text{FT}} = 0.71(2)$ for
 175 broadband excitation — a relative enhancement of 25 % (Figure 2c). These
 176 values are consistent with the spectral coherence protection mechanism es-
 177 tablished for the 7-site FMO complex [25]. The coherence lifetime τ_c , defined
 178 as the $1/e$ decay time of the l_1 -norm coherence $C_{l_1}(t) = \sum_{i \neq j} |\rho_{ij}(t)|$ [36], ex-
 179 tends from 280 ± 25 fs (broadband) to 420 ± 35 fs (filtered) — a 50 % increase
 180 that is causally linked to the yield gain: the extended coherence suppresses
 181 the pure-dephasing rate, lengthening the effective transport mean-free path
 182 and thereby increasing the fraction of the excitation wavepacket that reaches
 183 the reaction center before decohering. The mechanism is identical to that es-
 184 tablished for the 7-site system [25]: the dual-band filter selectively populates
 185 polaron-dressed states whose residual reorganization energy λ_{res} is reduced
 186 by 20 % relative to the full bath, suppressing pure dephasing (Section S3).

187 *NPoM cavity volume dependence of exciton transport*

188 The nanoparticle-on-mirror cavity couples the FMO complex to the OPV
 189 active layer through the strong coherent interaction described above, but the
 190 plasmonic mode also acts as an additional population decay channel that
 191 competes with the reaction center trapping. To quantify this trade-off, we
 192 perform a sensitivity scan of the NPoM mode volume V_{mode} , which controls
 193 the single-emitter coupling strength via $g_0 \propto V_{\text{mode}}^{-1/2}$ (Equation (2)). Smaller
 194 mode volumes produce stronger plasmon–exciton coupling but also enhance
 195 the plasmon’s role as a population sink, diverting excitation away from the
 196 reaction center.

197 Table 2 summarizes the forward transfer yield across seven mode volumes
 198 spanning the experimentally accessible sub-nanometer range [29]. The yield
 199 increases monotonically with mode volume, from $\Phi_{\text{FT}} = 0.0505$ at $V_{\text{mode}} =$
 200 0.2 nm^3 to a maximum of $\Phi_{\text{FT}} = 0.0799$ at $V_{\text{mode}} = 1.2 \text{ nm}^3$, with a slight
 201 decrease at 1.4 nm^3 . The trend is logarithmic with diminishing returns above
 202 1.0 nm^3 ; increasing the volume beyond 1.0 nm^3 recovers only 1 % of the
 203 baseline yield.

Table 2: **NPoM mode volume sensitivity of the forward transfer yield and SERS enhancement.** Baseline (no NPoM): $\Phi_{\text{FT}} = 0.98$. All NPoM runs with $L = 8$, $K = 2$, $\Delta t = 0.2$ fs, $t_{\text{max}} = 1$ ps. The $V_{\text{mode}} = 1.2 \text{ nm}^3$ entry is validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799 \pm 0.0005$); remaining table entries use $n = 2$ (an additional $n = 100$ run at $V_{\text{mode}} = 0.8 \text{ nm}^3$ is discussed in the text). SERS enhancement $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$ via Purcell factor scaling, normalized to $\text{EF}_{\text{SERS}} = 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$ [29]. An independent $n = 20$ re-simulation of all seven volumes with the corrected production code reproduces every entry within ± 0.002 ($V_{\text{mode}} = 0.8 \text{ nm}^3$ and 1.2 nm^3 exactly), cross-validating the low- n screening values.

$V_{\text{mode}} \text{ (nm}^3\text{)}$	Φ_{FT}	Δ vs. baseline (%)	EF_{SERS}
0.2	0.050 5	−94.8	1 600
0.4	0.060 5	−93.8	400
0.6	0.072 7	−92.6	180
0.8	0.076 8	−92.2	100
1.0	0.079 3	−91.9	64
1.2	0.079 9	−91.8	44
1.4	0.079 1	−91.9	33

204 The physical origin of this suppression is the strong-coupling-induced hy-
205 bridization of excitonic and plasmonic states. The dressed eigenstates acquire
206 a finite plasmonic character, which decays via the fast plasmon relaxation
207 pathway ($\kappa \approx 800 \text{ cm}^{-1}$, $Q \approx 15$) rather than populating the reaction center.
208 At $V_{\text{mode}} = 0.2 \text{ nm}^3$, where g_0 is largest, the plasmonic admixture in the
209 lower polariton branch reaches $\sim 15\%$, diverting a corresponding fraction of
210 the excitation flux. As V_{mode} increases, g_0 decreases, the polariton splitting
211 narrows, and the excitonic character of the bright states is partially restored.

212 Despite the monotonic recovery, all NPoM-on yields remain suppressed
213 by more than 90 % relative to the pristine FMO baseline ($\Phi_{\text{FT}} = 0.98$). This
214 indicates that, in the strong collective coupling regime accessible to NPoM
215 cavities ($g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$), the plasmonic decay pathway fundamentally
216 limits the maximum achievable transport efficiency. The SERS enhance-
217 ment factor ($\sim 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$) scales as $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$,
218 following the Purcell enhancement of the local field at both the excitation
219 and emission wavelengths. Table 2 includes the calculated SERS enhance-
220 ment for each mode volume, revealing a four-order-of-magnitude range (33 to
221 1.6×10^3) across the scanned volumes. The optimal volume represents a com-
222 promise: $V_{\text{mode}} \approx 1.2 \text{ nm}^3$ maximizes transport yield ($\Phi_{\text{FT}} = 0.0799$) while

retaining SERS sensitivity ($\text{EF}_{\text{SERS}} \approx 44$), sufficient for detecting 180 cm^{-1} Franck–Condon mode populations at the single-FMO-complex level [10]. By contrast, $V_{\text{mode}} = 0.2 \text{ nm}^3$ maximizes SERS sensitivity (1.6×10^3) at the cost of the lowest transport yield ($\Phi_{\text{FT}} = 0.0505$), a trade-off better suited to diagnostic-only applications where photochemical efficiency is secondary. At the reference volume, a dedicated $n = 100$ production run at $V_{\text{mode}} = 0.8 \text{ nm}^3$ confirms the same limiting behaviour ($\Phi_{\text{FT}} = 0.0768$, final plasmon population 0.79; [Section S11](#)).

Second, reducing the temperature to 77 K (analyzed as a theoretical limit-case for thermal phonon suppression in [Section S11.2](#)) doubles the trapping yield to $\Phi_{\text{FT}} = 0.1685$. In the main agrivoltaic operating regime at 295 K, the maximum trapped population on sites 3+4 remains at 3.8×10^{-3} for $V_{\text{mode}} = 1.2 \text{ nm}^3$, requiring the multi-scale sentinel fraction strategy to bypass plasmonic loading.

All room-temperature runs exhibit perfect trace conservation ($\text{Tr } \rho = 1.0000$), validating the numerical stability of the PT-HOPS/SBD propagator. The $V = 0.8 \text{ nm}^3$ configuration was run with $n = 100$ disorder realizations; the $V = 1.2 \text{ nm}^3$ configuration was validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799$, within 0.6 % of the $n = 2$ estimate of 0.0804). The baseline NPoM-off run ($n = 20$) is shown as a limiting-case reference; quantitative comparison across configurations is limited by the differing trajectory counts.

Global canopy yield integration

Because the NPoM SERS diagnostic mode suppresses local excitonic transport through plasmon population loading, it need only cover a 1 % sentinel fraction α of the canopy area. The remaining 99 % operates in passive OPV mode, preserving the near-baseline forward transfer yield $\Phi_{\text{FT}}^{\text{passive}} = 0.980$. The area-weighted *global* canopy trapping yield is therefore:

$$\Phi_{\text{FT}}^{\text{global}} = \alpha \Phi_{\text{FT}}^{\text{NPoM}} + (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}}, \quad (5)$$

yielding $\Phi_{\text{FT}}^{\text{global}} = 0.971$ at $\alpha = 0.01$ and $\Phi_{\text{FT}}^{\text{NPoM}} = 0.080$ (or 0.0799 precisely); canopy-scale productivity is thus maintained even though transport is suppressed locally.

Dynamic photo-protection via Floquet–Stark detuning and OMIT

Under high solar flux ($> 800 \text{ W m}^{-2}$), the OPV dielectric constant changes dynamically, inducing a time-dependent Stark shift of the excitonic

256 levels that detunes the OPV–FMO coupling. We model this as a Floquet
257 driving term:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_F t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (6)$$

258 with amplitude $V_0 = 0.05 \text{ eV}$ and frequency $\omega_F = 1.2 \text{ THz}$. When the in-
259 cident flux exceeds the threshold, the time-dependent detuning shifts the
260 OPV exciton energies out of resonance with the FMO absorption bands, re-
261 ducing energy transfer to the biological complex by up to 40 % (Figure 3a).
262 To prevent rapid, noise-induced oscillations under fluctuating cloud cover,
263 a hysteresis buffer of $\pm 50 \text{ W m}^{-2}$ is implemented (activating the detuning
264 switch at 850 W m^{-2} and deactivating it only when the flux drops below
265 750 W m^{-2}).

266 Concurrently, an optomechanically induced transparency (OMIT) mecha-
267 nism [37] in the NPoM cavity creates destructive interference that suppresses
268 transmission at the 750 nm and 820 nm channels under high flux. The trans-
269 mission is modulated as:

$$T_{\text{OMIT}}(I) = T_0 \left(1 + \frac{I}{I_{\text{sat}}} \right)^{-1}, \quad (7)$$

270 where I is the incident flux and $I_{\text{sat}} = 800 \text{ W m}^{-2}$. At $1\,000 \text{ W m}^{-2}$, the trans-
271 mission drops to 44 % of baseline (Figure 3b), providing dynamic protection
272 against exciton overload. At zero flux (night mode), both mechanisms return
273 to zero without singularities (Section S4).

274 *In situ SERS optomechanical diagnostics and environmental robustness*

275 The NPoM platform enables non-destructive readout of FMO vibrational
276 states via surface-enhanced Raman scattering (SERS) [38]. Recent scoping
277 reviews of in-situ agrivoltaic sensing list no SERS or plasmonic category [39],
278 which is the integration gap addressed here. Multilayer graphene–metal
279 plasmonic architectures have demonstrated detection sensitivities exceeding
280 conventional SPR biosensors by an order of magnitude [40]. The optome-
281 chanical coupling between the plasmon mode and FMO molecular vibrations
282 ($g_{\text{om}} = 0.01$) produces a SERS enhancement factor of $\text{EF}_{\text{SERS}} = 44$ at the
283 optimal $V_{\text{mode}} = 1.2 \text{ nm}^3$, reaching 1.6×10^3 for the smallest scanned mode
284 volumes.

285 Beyond the three canonical BChl *a* vibrational modes previously iden-
286 tified (180 cm^{-1} , 740 cm^{-1} , $1\,145 \text{ cm}^{-1}$), we here assign four agriculturally
287 specific target signatures to the sentinel sub-system. **Oxidative stress** is

288 read out through the C–C stretch of conjugated chlorophyll *a* at $1\,145\,\text{cm}^{-1}$,
 289 with a detection limit of $50\,\text{nmol L}^{-1}$ in the pre-necrotic regime [10]. **Pes-**
 290 **ticide contamination** by 2,4,5-trichlorophenoxyacetic acid (2,4,5-T) is de-
 291 tected via its characteristic Raman peak at $1\,435\,\text{cm}^{-1}$ ($\text{LOD} = 1\,\text{nmol L}^{-1}$
 292 under controlled conditions), leveraging the plasmonic hot-spot amplification
 293 in the NPoM gap [10]. **Heavy metal contamination** of irrigation water
 294 (Pb^{2+}) is quantified through Stern–Volmer fluorescence quenching of encaps-
 295 ulated CdTe/ZnSe quantum dots ($\text{LOD} = 31.8\,\text{nmol L}^{-1}$ under controlled
 296 conditions) [41] sealed within the NPoM sensor probe polymer matrix, which
 297 are immune to the dust and biofouling that affect bare gold surfaces. Cross-
 298 sensitivity between soil moisture and Pb^{2+} contamination is resolved by dual
 299 spectral interrogation: heavy metals modify the Stern–Volmer quenching in-
 300 tensity and fluorescence decay time, whereas soil moisture alters the local
 301 dielectric constant, producing a spectral broadening that is spectrally or-
 302 thogonal to the metal-induced quenching signature. **Pathogen detection**
 303 is achieved via Nitrogen-Vacancy (NV) center nanodiamond sensors operated
 304 in a microfluidic “Point-of-Care” (Lab-on-a-Chip) desktop reader located at
 305 the greenhouse hub (Section S12.7). These solid-state quantum sensors de-
 306 tect paramagnetic bacterial metabolites from sap or irrigation water samples
 307 at pre-symptomatic stages ($\text{LOD} = 0.8\,\text{nmol L}^{-1}$ under controlled conditions;
 308 the agricultural matrix will degrade in-field sensitivity, which is mitigated by
 309 the graphene barrier and *DynamicCalibrator* corrections).

310 Figure 3c shows the simulated Raman spectrum for the three BChl *a*
 311 modes under active NPoM coupling. The relative intensities encode the in-
 312 stantaneous exciton population distribution, providing a direct optical probe
 313 of the energy transfer pathway.

314 **Environmental robustness.** The realistic agricultural environment
 315 (dust, bird droppings, microbial fouling) is modeled at three infrastructure
 316 levels. At the OPV panel level, the soiling factor $\eta_{\text{soil}}(t)$ decays exponentially
 317 with days since last cleaning:

$$\eta_{\text{soil}}(t) = \max(1 - 0.005 t_{\text{days}}, 0.75), \quad (8)$$

318 which limits maximum OPV efficiency loss to 25 % at the worst-case cleaning
 319 interval. At the sensor level, the graphene monolayer spacer of each NPoM
 320 cavity is functionalized with a zwitterionic polymer coating [42, 43], modeled
 321 in our simulations to preserve $\geq 90\%$ quantum coherence after 60 d continu-
 322 ous field immersion; this coating constitutes the primary anti-fouling barrier

that physically repels microbial adhesion and protein adsorption. A thin-film micro-shielding layer (1 μm to 5 μm of SiO_2 or Al_2O_3) around the NPoM picocavity and QCD sensor tip further attenuates the diurnal temperature excursion experienced by the quantum probe, attenuating the thermal-drift contribution by the factor $(1 - \eta_{\text{shield}})$, with $\eta_{\text{shield}} = 0.85$ (Section S12.5). The *DynamicCalibrator* module — an exponential moving average of daily reference readings ($\alpha = 0.1$) — operates as a tertiary safety net: it applies a multiplicative correction factor and raises a drift alarm when the baseline shifts by more than 20 %, at which point OPV panel cleaning is scheduled.

Smart shield microclimate: FAO-56 evapotranspiration

The spectrally selective OPV layer functions as a “smart shield” that blocks ultraviolet and near-infrared radiation while transmitting photosynthetically active bands. Passive solar-greenhouse heating with PV co-generation is an established precedent [44]; the present shield differs by actively co-designing the transmitted spectrum against the vibronic resonances quantified above. We parameterize the greenhouse microclimate using the FAO-56 Penman–Monteith equation [30] adapted for shaded conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (9)$$

where R_n is the net radiation attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$ (with soil heat flux $G \approx 0$ assumed for daily timescales under greenhouse shielding), Δ is the slope of the saturation vapor pressure curve, γ is the psychrometric constant, u is wind speed, $e_s - e_a$ is the vapor pressure deficit, and $K_c = 0.85$ is the crop coefficient.

Under standard conditions (600 W m^{-2} , 25 $^\circ\text{C}$, 60 % humidity, 2 m s^{-1} wind), the open-field evapotranspiration is 4.5 mm d^{-1} . With the OPV smart shield, ET_c drops to 3.2 mm d^{-1} — a 28 % reduction — translating to 1.3 mm d^{-1} in water savings (Figure 4a). The crop coefficient $K_c = 0.85$ accounts for the reduced transpiration demand under partial shade [30].

Socioeconomic viability and cooperative payback

Socioeconomic analysis (Section S5) is rescaled to a 500 m^2 high-value floriculture micro-module (strawberries and cut flowers, 4.5 USD kg^{-1} at 8.2 $\text{kg m}^{-2} \text{yr}^{-1}$), which reduces the entry-level capital expenditure to 161 000 USD (at 322 USD m^{-2}). Our $\$/\text{kWh}$ accounting follows the

356 Monte-Carlo techno-economic logic established for European agrivoltaic
 357 configurations [45]. A blended-finance subsidy of 30 % reduces the net
 358 CAPEX to 112 700 USD.

359 Three operating expenditure streams are modeled: extension services and
 360 technician training (1 200 USD yr⁻¹), OPV panel cleaning (800 USD yr⁻¹),
 361 and general maintenance (2 000 USD yr⁻¹), for a total annual OPEX
 362 of 4 000 USD yr⁻¹. The net annual cash flow of 26 612 USD yr⁻¹ (from
 363 30 612 USD yr⁻¹ total revenue minus 4 000 USD of OPEX) yields a sim-
 364 ple payback period of 4.23 yr (or 6.05 yr without subsidy) and a 10-year
 365 discounted NPV of +37 664 USD at a 12 % risk-adjusted discount rate
 366 appropriate for Sub-Saharan Africa [46] (Figure 4c). Holding other inputs
 367 fixed, varying the subsidized net CAPEX by $\pm 20\%$ shifts the payback
 368 to 3.39–5.08 yr; a 50 % higher OPEX (6 000 USD yr⁻¹) gives 4.58 yr, and
 369 20 % lower revenue gives 5.50 yr. While a baseline CAPEX of 322 m⁻²
 370 represents a significant barrier in Sub-Saharan Africa relative to local
 371 income levels, utilizing local construction materials (such as bamboo and
 372 timber framing instead of structural steel) and green chemical synthesis of
 373 gold nanoparticles from local plant extracts can reduce the capital cost to
 374 140 m⁻², lowering the unsubsidized payback period to 2.63 yr and making
 375 the modelled system potentially viable for local cooperatives under these
 376 cost assumptions. By contrast, the remaining barriers are physical rather
 377 than economic: fabrication of sub-2 nm NPoM gaps still requires laboratory-
 378 grade nanofabrication and a stable Au/graphene supply chain, and OPV
 379 encapsulation lifetime under tropical irradiance is the binding reliability
 380 constraint (Section S12.5). Subsidy policy alone cannot clear these barriers;
 381 they define the process-engineering roadmap that separates this modelled
 382 proof of concept from field deployment. Taken together, these sensitivities
 383 demonstrate that high-value crop selection and blended-finance structurally
 384 transform the investment case.

385 *Secure IoT telemetry via BB84 quantum key distribution*

386 Field-to-cloud telemetry is secured using a simulated BB84 quantum key
 387 distribution protocol [31] over a buried single-mode fiber link, which elimi-
 388 nates the atmospheric turbulence and scintillation that affect free-space chan-
 389 nels in agricultural environments. This quantum-secured telemetry layer is
 390 physically integrated with the NPoM-CQD sentinel network: because the
 391 high-sensitivity biological and pesticide diagnostics control high-stakes ir-
 392 rigation and crop-protection actuators (which could be targeted by hostile

actors to cause artificial crop failure or pesticide contamination), securing these feedback commands with BB84 QKD protects the command integrity of the agrivoltaic micro-grid. Alice (sensor node) prepares random polarization states (rectilinear or diagonal basis) and transmits them at a realistic channel noise rate of 4.8 % QBER [47]. Bob (cloud server) measures in random bases, and the sifted key is extracted from matching-basis events. The QBER is estimated on a sample fraction; if QBER exceeds the Shor–Preskill unconditional-security threshold of 11 % [48], the protocol raises a **SecurityThresholdExceeded** exception that halts all QKD-protected irrigation commands, preventing data-injection attacks under eavesdropping.

With a nominal fiber noise rate of 4.8 %, the protocol achieves QBER = 4.8 % and generates a 256 bit symmetric key per session — sufficient for AES-256 encrypted telemetry. Field trials of BB84 using deterministic single-photon sources have demonstrated stable key generation over installed fiber links [47], supporting the practical feasibility of this approach under comparable link budgets. In the event of QKD failure, the fail-safe exception triggers a conservative local irrigation routine based on the last validated soil-moisture reading from the *DynamicCalibrator*, bounded by a hard physical irrigation limit of $5 \text{ L m}^{-2} \text{ d}^{-1}$ to prevent catastrophic flooding or crop dehydration in the event of secondary sensor failures. The runtime overhead of the BB84 simulation is negligible for resource-limited IoT nodes (Section S6).

Experimental testability

The modelled predictions rest on parameters that are independently measurable (Table 3). Two protocols can validate them directly. First, two-dimensional electronic spectroscopy (2DES) at 295 K with spatial-light-modulator pulse shaping into dual-band channels at 750 nm and 820 nm can track the diagonal and cross-peak quantum-beat oscillations predicted here, verifying the extension of τ_c from 280 fs to 420 fs [49, 16]. Second, time-resolved SERS in the NPoM geometry can monitor the 180 cm^{-1} and 1145 cm^{-1} BChl *a* modes in real time and quantify the area-weighted yield $\Phi_{\text{FT}}^{\text{global}} = 0.971$ under ambient conditions [29, 50]. More broadly, recent assessments identify quantum sensing for precision agriculture as an emerging opportunity whose open calibration and robustness gaps match the calibration architecture proposed here [12].

Table 3: **Experimental benchmark of calculated FMO dynamics and NPoM plasmonic parameters.** Calculated at $T = 295$ K, $L = 8$, $K = 2$.

Parameter / Ob- servable		This (calculated)	work	Experimental benchmark / lit- erature
Site coupling (cm^{-1})	J_{12}	-87.7		-87.7 ± 5.0 (2DES, Brixner 2005 [49])
Reorganization energy λ_D (cm^{-1})		35		35–40 (2DES, Engel 2007 [15])
Cutoff frequency (cm^{-1})	γ_D	50		50 (2DES, Panitchayangkoon 2010 [16])
Coherence lifetime (passive)	τ_c	420 ± 35 fs		400 ± 50 fs (2DES room temp., Pan- itchayangkoon 2010 [16])
NPoM coupling ($V_{\text{mode}} = 1.2 \text{ nm}^3$)	g_0	110 cm^{-1}		$120 \pm 15 \text{ cm}^{-1}$ (NPoM, Chikkaraddy 2016 [29])
SERS EF _{SERS}	enhancement	44		10^2 – 10^3 (sub-nm NPoM, Baumberg 2019 [50])

Discussion

Our results demonstrate that quantum coherence engineering at the bio-organic interface can deliver modelled sustainability benefits across microscopic energy transfer, crop water use, carbon footprint, and data security.

Mechanism generality. The dual-band filtering effect is not specific to the FMO complex. The resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$ (Section S7) is a general property of vibronically coupled chromophore aggregates [17]. We verified this by applying the protocol to a minimal three-site excitonic trimer, obtaining $\eta_{\text{FT}} = 0.18 \pm 0.03$ (Section S8), qualitatively consistent with the quantum phase synchronization mechanism identified in photosynthetic antennas [28]. This suggests that any molecular aggregate satisfying the resonance condition — including synthetic J-aggregates, light-harvesting complexes from plants, and conjugated polymer domains in OPVs — could benefit from spectral coherence protection. Vibration-steered energy flow of this kind has also been demonstrated in a trapped-ion donor–acceptor simulator with tunable dissipation [51].

The “quantum divide.” The advanced materials required for quantum agrivoltaics — graphene, plasmonic nanoparticles, OPVs with engineered bandgaps — are not yet accessible to subsistence farmers in developing

economies. Our cooperative cost-sharing model demonstrates that with a carbon-linked subsidy of 30 % and a cooperative of 5 members, the payback period drops below 4 yr, making the technology economically viable under the modelled subsidy. We propose a tiered subsidy framework where the NEB itself serves as the verifiable metric for subsidy qualification, linking quantum yield improvements to farmer access.

Quantum digital twin. The modular architecture — quantum interface, FAO-56 microclimate, LCA, IoT security — serves as the physics engine of an agrivoltaic *digital twin*: a model-based supervisory replica that combines sensor telemetry from the *DynamicCalibrator* with the PT-HOPS dynamics engine to recommend irrigation schedules, OPV cleaning dates (from soiling factor [Eq. \(8\)](#)), and carbon credit yields. This device-physics integration is distinct from the supply-chain and breeding applications mapped in recent quantum-in-agriculture reviews [\[11\]](#).

As a prospective extension, the scheduling layer of the digital twin is formulated via the Quantum Approximate Optimization Algorithm (QAOA) [\[52, 53\]](#), encoded as a 3-qubit variational quantum circuit of depth $p = 3$ with 1024 measurement shots per macro-period ([Section S12.3](#)). The algorithm encodes the irrigation–refrigeration–grid-export trilemma as a quadratic unconstrained binary optimization (QUBO) cost Hamiltonian and solves it via a PennyLane emulator at sub-hourly timescales. Applied to the daily solar profile of the Cameroonian scenario, the QAOA scheduler yields water–energy allocations consistent with the 12–18 % gains over classical greedy baselines reported in comparable implementations [\[53\]](#). As a further prospective readout, our `quantum_gravimetry.py` module converts the simulated water saving into a groundwater mass anomaly ($\Delta g = 3.5 \text{ nm s}^{-2}$ for a single greenhouse — below single-instrument noise but potentially detectable with a small regional gravimeter network or gradiometric common-mode rejection), with the full noise-budget and inversion discussion given in [Section S12.2](#) [\[54, 55\]](#).

A hybrid quantum-classical signal processing pipeline — combining Matrix Product State (MPS) denoising with quantum kernel ridge regression (QKRR) [\[56, 57\]](#) — further enables the digital twin to extract pre-symptomatic stress signatures from the noisy SERS and CQD fluorescence telemetry. To satisfy edge computation constraints, the heavy MPS tensor contraction and QAOA optimization are executed on the centralized cooperative server (or cloud), with the resulting optimized weights transmitted back to the field nodes — an edge–cloud split complementary to classical

drift-diffusion device simulators [58]. The local ARM Cortex-M4 IoT node executes only the low-latency inference of the pre-trained, frozen-weight QKRR model. The MPS compressor acts as a nonlinear low-pass filter on the server that suppresses greenhouse noise (thermal fluctuations, biofouling, dust) while preserving the sharp vibronic features at 180 cm^{-1} and 1145 cm^{-1} that classical Savitzky–Golay filters tend to smear (Section S10). The cleaned spectra are projected through the quantum-inspired kernel for anomaly scoring, issuing early warnings that allow preemptive agronomic intervention before macroscopic symptoms appear — all within a 3.7 ms local compute budget on the IoT node. We emphasize that this sub-millisecond execution latency is not designed to match the slower multi-hour physical response times of soil moisture percolation and plant transpiration. Rather, it enables spatial multiplexing: a single ARM Cortex-M4 edge processor can sequentially query, compress, and classify SERS telemetry from dozens of distributed NPoM sensor nodes across the 500 m^2 greenhouse in under a second, reducing the electronic-component count. This chain connects microscopic quantum efficiency to field-level agronomic management; no existing agrivoltaic design integrates both [46, 6, 2].

Prior art and positioning. Previous agrivoltaic studies have focused on light-sharing through passive spectral filtering [6] or crop selection [2]. Semi-transparent OPVs with power conversion efficiencies exceeding 15 % and 40 % average visible transmittance have recently been demonstrated through blade-coating scale-up methods [59], and complementary work on semitransparent organic photovoltaics for greenhouse applications has shown power conversion efficiencies above 12 % [7]. Our contribution is to add an active, quantum-coherent dimension to the spectral filter: the OPV is no longer a passive absorber but an active spectral shaper that prepares coherence-protected states in the photosynthetic apparatus. Multi-vector solar agri-systems combining agrivoltaics with hydrogen and desalination set the *Applied Energy* bar for water–food–energy integration [60]; we differentiate by quantum-guided spectral co-design with integrated sensing, and we adopt the functional-unit discipline (emissions per kWh, economics per configuration) of recent agrivoltaic TEA/LCA studies [8, 9]. Relative to this literature, the present work provides, to our knowledge, the first (i) agrivoltaic stack whose spectral architecture is co-designed by numerically exact open-quantum-system dynamics, jointly optimizing excitation transport and power conversion; (ii) monolithic photothermal–PV–SERS device concept with field-relevant calibration in a single footprint; and (iii) multifunctional

522 techno-economic and lifecycle framework that internalizes heat, crop, water,
523 sensing, and data-security trade-offs in $\$/W + \$/\text{kg} + \$/\text{m}^3$ terms. [Ta-](#)
524 [ble 4](#) condenses this positioning: each prior class delivers two to three of the
525 seven functions, whereas the present stack co-optimizes all seven in a single
526 footprint under one accounting.

Table 4: **Capability matrix of agrivoltaic and agri-sensing approaches.** $\checkmark$ = function delivered within the stack; $-$ = partial or incidental; $—$ = absent. Classes: passive spectral claddings [6, 2, 7]; multi-vector agri-energy systems (agrivoltaics coupled to hydrogen and desalination) [60]; standalone agri-sensing hardware [10, 11, 12]. Multifunctional accounting denotes joint $\$/W + \$/\text{kg} + \$/\text{m}^3$ techno-economic and lifecycle evaluation of all functions. No prior class delivers more than three of the seven functions; the present stack delivers all seven in a single footprint with unified accounting.

Function	Passive	Multi-vector	Sensing	This work
Electrical power (OPV)	$\checkmark$	$\checkmark$	$—$	$\checkmark$
Crop microclimate	$\checkmark$	$\checkmark$	$—$	$\checkmark$
Water management	$-$	$\checkmark$	$—$	$\checkmark$
Photothermal heat	$—$	$-$	$—$	$\checkmark$
In-situ SERS sensing	$—$	$—$	$\checkmark$	$\checkmark$
QKD-secured telemetry	$—$	$—$	$—$	$\checkmark$
Multifunctional accounting	$—$	$—$	$—$	$\checkmark$

527 This approach is distinct from prior quantum biology studies [61] that
528 treat the environment as a fixed noise source; here, the photonic environ-
529 ment itself is engineered through the NPoM cavity to maximize coherence
530 protection. A key design choice warrants explicit justification: the NPoM
531 cavity, despite imposing a $> 90\%$ penalty on excitonic transport ([Table 2](#)),
532 is retained because its *diagnostic function* is dedicated — it enables non-
533 destructive, high-sensitivity SERS readout of the FMO vibrational state,
534 pesticide levels, and heavy-metal contamination. The transport penalty is
535 absorbed by confining NPoM operation to the 1% sentinel fraction ([Equa-](#)
536 [tion \(5\)](#)), preserving 97.1% of the global canopy trapping yield while main-
537 taining full diagnostic capability.

538 **Limitations.** Several approximations constrain our analysis. The quan-
539 tum dynamics are restricted to the single-exciton manifold; under high solar
540 flux, multi-exciton states, singlet fission, or exciton-exciton annihilation in
541 the OPV and FMO interfaces could alter the collective coupling strength
542 g_{eff} and transport efficiency. The bath is treated as site-independent, and

543 the reaction center trapping is modeled as a phenomenological Lindblad-like
 544 rate. The FAO-56 model, while standard, does not capture transient stom-
 545 atal responses to fluctuating light or the dynamic microclimatic feedback
 546 between canopy transpiration and vapor pressure deficit (VPD) inside the
 547 greenhouse envelope. Sensitivity analysis over plausible wind-speed varia-
 548 tions ($u = 0.2 - -0.8 \text{ m s}^{-1}$) yields a $\pm 10\%$ uncertainty band on the predicted
 549 28 % water saving, which propagates into a $\pm 10\%$ range on the annual irri-
 550 gation volume and corresponding NEB carbon offset. The BB84 simulation
 551 assumes a buried single-mode fiber link; while buried fibers are vulnerable to
 552 agricultural mechanical damage (plowing, root intrusion), a free-space opti-
 553 cal (FSO) link would suffer from strong scattering under dusty greenhouse
 554 environments, requiring dedicated atmospheric correction models. While
 555 software-based post-quantum cryptography offers a lower-cost alternative,
 556 the physical BB84 protocol provides information-theoretic security immune
 557 to algorithmic cryptanalysis on resource-constrained embedded nodes such
 558 as the ARM Cortex-M4 irrigation controllers. The 1 % sentinel fraction is a
 559 design parameter; field optimization will require empirical calibration of the
 560 SERS LODs under actual fouling conditions. We conservatively estimate a
 561 $\times 10$ matrix-effect penalty on all reported LODs when transitioning from con-
 562 trolled laboratory buffers to raw agricultural sap or irrigation water, owing
 563 to competitive binding, background fluorescence, and ionic strength effects;
 564 the *DynamicCalibrator* compensates for this attenuation through real-time
 565 baseline correction. The zwitterionic polymer coating modeled on the NPoM
 566 graphene spacer ([Section S12.1](#)) acts as the primary fouling barrier in our
 567 simulations, with the *DynamicCalibrator* serving as a secondary correction
 568 layer; the dual architecture reduces the mandatory cleaning frequency, which
 569 is already factored into the cooperative OPEX model. We stress that the 60-
 570 day resilience of zwitterionic coatings applies strictly to sensors sheltered in
 571 protected micro-sampling chambers or rigid ceramic encasings; direct, un-
 572 protected exposure to abrasive soil matrices (bioturbation, quartz particles,
 573 tillage) would lead to mechanical scraping and rapid surface degradation.

574 The 30 % blended-finance subsidy required to achieve a 4.23 yr payback
 575 period cannot be bridged by voluntary carbon credits alone (generating only
 576 98 USD yr^{-1} at $10 \text{ USD t}_{\text{CO}_2\text{e}}^{-1}$ for the 500 m^2 module); it must be underpinned
 577 by dedicated rural-adaptation finance channels such as the Green Climate
 578 Fund (GCF) and the African Development Bank’s Climate Change Fund
 579 (Fonds Spécial Climat-Dev de la BAD), which are mandated to co-finance
 580 climate-resilient agriculture in Sub-Saharan Africa.

581 **Outlook.** The integration of quantum coherence engineering with agri-
582 cultural sustainability suggests an axis for agrivoltaic design. Three experi-
583 mental and theoretical directions are prioritized for future work.

584 First, on the hardware and materials scale, the quantum model must
585 be extended to full multi-exciton dynamics under realistic solar spectra, ac-
586 companied by experimental validation via two-dimensional electronic spec-
587 troscopy (2DES) with spatial light modulator pulse shaping [62, 63]. Field tri-
588 als of graphene-based NPoM coatings on greenhouse cladding with the CQD
589 sentinel network are required to evaluate in-situ sensor performance under
590 actual biofouling conditions and verify the coherence-preservation lifetime
591 of the zwitterionic functionalization layer (Section S12.1). Alternative low-
592 loss plasmonic architectures — dielectric nanoantennas (Si, TiO₂) support-
593 ing Mie resonances [64] and nanocavity-free substrates — should be explored
594 to minimize the NPoM transport penalty while preserving SERS sensitivity.
595 Additional hardware extensions include passive thermal micro-shielding (Sec-
596 tion S12.5), UiO-66 Zr-based quantum metal-organic frameworks for coherent
597 contaminant remediation (Section S12.6), and quantum fertilizer biostimu-
598 lation nanoparticles to enhance chlorophyll yields (Section S12.8).

599 Second, scaling the digital twin to a multi-cooperative network will re-
600 quire hardware-in-the-loop QKD validation under dusty field conditions and
601 deployment of a real-time aquifer gravimetric array to confirm that the mod-
602 eled groundwater recharge connects at the satellite geodesy scale. Monte
603 Carlo uncertainty propagation in the full LCA should be extended with cor-
604 related climatic input distributions drawn from regional reanalysis data.

605 Third, to prevent these advances from widening the “quantum divide,”
606 we have defined the Global Quantum Agriculture Standard (GQAS), a six-
607 pillar certification framework covering QKD security, data sovereignty, sen-
608 sor detection limits, quantum fidelity, latency, and lifecycle sustainability
609 (Section S12.9). Our Cameroon cooperative proof-of-concept scenario serves
610 as the first application of this proposed framework, satisfying all six GQAS
611 criteria and demonstrating internal consistency across the multi-domain sim-
612 ulation. Independent experimental validation will be required before GQAS
613 can be adopted as an operational standard. Scaling GQAS adoption to small-
614 holder networks across Sub-Saharan Africa and Southeast Asia requires subsi-
615 dized sensor supply chains, open-access certification auditing, and regulatory
616 recognition on premium export markets such as GlobalGAP.

617 Conclusions

618 We asked whether exact non-Markovian quantum dynamics can jointly
 619 optimize excitation-transport and power-conversion efficiency in a sensing-
 620 integrated photothermal agrivoltaic stack, with a positive multifunctional
 621 net benefit under field constraints. Within the modelled 500 m^2 cooper-
 622 ative case, the answer is affirmative but conditional: dual-band filtering
 623 raises the forward transfer yield by 25 % and extends coherence lifetime
 624 by 50 %; confining NPoM operation to a 1 % sentinel fraction preserves
 625 97.1 % global canopy yield while retaining SERS diagnostics; the FAO-56
 626 smart shield cuts evapotranspiration by 28 % with a Net Ecological Ben-
 627 efit of $19.6\text{ kg}_{\text{CO}_2\text{e}}\text{ m}^{-2}\text{ yr}^{-1}$; and the cooperative case pays back in 4.23 yr
 628 at 30 % subsidy. These claims are bounded by the $> 90\%$ local NPoM
 629 transport penalty, the single-exciton and site-independent-bath approxima-
 630 tions, the $\pm 10\%$ microclimate uncertainty band, and the modelled (not field-
 631 validated) fouling and matrix-penalty assumptions. For deployment, the eco-
 632 nomic barrier is dominant and tractable — local-material construction and
 633 green-synthesis gold lower the capital cost from 322 USD m^{-2} to 140 USD m^{-2}
 634 (unsubsidized payback 2.63 yr) — whereas the physical barriers (nanofabri-
 635 cation of sub-2 nm NPoM gaps, in-field fouling of the sentinel fraction, OPV
 636 encapsulation lifetime under tropical irradiance) require process engineering
 637 rather than subsidies. Priority next steps are 2DES validation of the fil-
 638 ter protocol, low-loss dielectric antenna alternatives, field fouling trials of
 639 the sentinel network, and a localized Au/graphene sensor supply chain for
 640 Sub-Saharan manufacturing.

641 Methods

642 *Proof-of-Concept scenario: 500 m² Floriculture Micro-Module*

643 To ground the multi-domain framework in a concrete operational context,
 644 we consider a 500 m^2 high-tunnel greenhouse located in Cameroon’s West-
 645 ern Highlands. This region supports high-value floriculture (strawberries, cut
 646 flowers) suitable for micro-module deployment by cooperatives. The environ-
 647 ment features a mean annual solar irradiance of $5.76\text{ kWh m}^{-2}\text{ d}^{-1}$ [65] and
 648 significant biofouling potential that necessitates the NPoM *DynamicCalibra-*
 649 *tor* and regular OPV cleaning (factored into the cooperative OPEX model).
 650 Under the NPoM-shield spectral filter, the 28 % reduction in crop evapotran-
 651 spiration (ET_c) relative to the open-field FAO-56 baseline corresponds to the

652 modelled water-savings credit of $1\,300\text{ L m}^{-2}\text{ yr}^{-1}$ ($650\text{ m}^3\text{ yr}^{-1}$ for the 500 m^2
 653 greenhouse; [Section S5](#)). The area-weighted global canopy trapping yield
 654 ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) ensures near-baseline productivity while accommodating a
 655 1% NPoM sentinel fraction for SERS diagnostics. The 15% quantum yield
 656 enhancement observed in the passive OPV fraction translates to an addi-
 657 tional $1.23\text{ kg m}^{-2}\text{ yr}^{-1}$ of high-value produce (strawberries at 4.5 USD kg^{-1}),
 658 generating substantial supplemental revenue. The OPV module generates 90
 659 MWh yr^{-1} of electrical energy for a 500 m^2 installation at $180\text{ kWh m}^{-2}\text{ yr}^{-1}$.
 660 Displacing grid electricity at $450\text{ g}_{\text{CO}_2\text{e}}\text{ kWh}^{-1}$ offsets $40.5\text{ t}_{\text{CO}_2\text{e}}\text{ yr}^{-1}$. Com-
 661 bined with the NEB ($19.6\text{ kg}_{\text{CO}_2\text{e}}\text{ m}^{-2}\text{ yr}^{-1}$, [Figure 4b](#)), the system achieves
 662 net-zero operational carbon rapidly. At $10\text{ USD t}_{\text{CO}_2\text{e}}^{-1}$ [66], the system gener-
 663 ates 98 USD yr^{-1} in verified carbon revenue. The 500 m^2 scale is well matched
 664 to cooperative ownership models: under a 30% blended-finance subsidy, the
 665 payback shortens from 6.05 yr unsubsidized to 4.23 yr, with each of the five
 666 cooperative members exposed to only 22 540 USD of net capital. This demon-
 667 strates that quantum-enhanced agrivoltaics can be economically viable under
 668 modelled assumptions when coupled with strategic crop selection and robust
 669 field-sensor integration.

670 *PT-HOPS/SBD quantum dynamics*

671 All quantum dynamics simulations use the Process Tensor Hierarchy of
 672 Pure States with Stochastically Bundled Dissipators as implemented in the
 673 open-source MesoHOPS library (v1.7) [20]. The hierarchy depth is $L = 8$
 674 with $K = 2$ Matsubara terms and a time step $\Delta t = 0.2\text{ fs}$, established as the
 675 production standard for converged dynamics at 295 K [25]. Static disorder
 676 is modeled as Gaussian fluctuations of the site energies with standard devi-
 677 ation $\sigma = 50\text{ cm}^{-1}$, sampled over $n = 100$ realizations. The bath spectral
 678 density follows the 12-mode Kleinekathöfer/Coker parameterization with a
 679 Drude–Lorentz background ($\lambda_D = 35\text{ cm}^{-1}$, $\gamma_D = 50\text{ cm}^{-1}$) and 12 under-
 680 damped vibronic modes (frequencies $180\text{--}1\,500\text{ cm}^{-1}$, Huang–Rhys factors
 681 $S_k = 0.003 - -0.050$).

682 *FAO-56 evapotranspiration*

683 Crop evapotranspiration is computed using the standard FAO-56
 684 Penman–Monteith equation [30] modified for greenhouse conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (10)$$

with R_n attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$. We note that the modified FAO-56 formulation represents a first-order approximation; under the enclosed high-tunnel envelope, boundary layer resistance and sensible heat exchange are altered relative to open field conditions. Future iterations of the digital twin should integrate three-dimensional Computational Fluid Dynamics (CFD) modeling to map localized microclimatic gradients and latent heat transfer. Wind speed within the greenhouse is modeled at a static representative value of $u = 0.5 \text{ m s}^{-1}$, corresponding to a typical 10 % reduction of average external wind velocities. Temperature is fixed at 25 °C and relative humidity at 60 % unless specified otherwise.

BB84 QKD simulation and fail-safe IoT architecture

The BB84 protocol is implemented as a Python simulation of polarization-state preparation, noisy transmission over a buried single-mode fiber channel, sifting, and QBER estimation. Alice prepares $4 \times N$ random bits (for $N = 256$ key length) in random rectilinear or diagonal bases. Bob measures in random bases, and matching-basis events are retained for key sifting. QBER is estimated on a sample fraction (minimum 1 bit, maximum 50 bits). If QBER exceeds the Shor–Preskill threshold of 11 % [48], a `SecurityThresholdExceeded` exception is raised, halting the QKD-protected irrigation command relay. A last-validated soil-moisture reading from the *DynamicCalibrator* is used to execute a conservative local irrigation fallback.

SERS enhancement scaling limits

The SERS enhancement factor $\text{EF}_{\text{SERS}} \approx 44$ at the optimal mode volume $V_{\text{mode}} = 1.2 \text{ nm}^3$ is computed using standard Purcell-factor scaling of the localized electromagnetic field. We note that this standard $|E|^4$ approximation represents a lower-bound semi-classical estimate. In sub-nanometer picocavities ($V_{\text{mode}} < 1.5 \text{ nm}^3$), local spatial dispersion, quantum tunneling, and non-local dielectric screening alter the field distribution relative to classical electrodynamics, while molecular optomechanical coupling can pump vibrational modes out of thermal equilibrium. Precise calculation of the localized fields in this regime requires non-local Green’s tensor integration, which motivates the proposed in-situ experimental validation. The benchmark against the experimentally reported 10^2 – 10^3 range for sub-nanometer NPoM gaps [50] is given in Table 3.

720 *DynamicCalibrator, thermal shielding, and soiling factor*

721 Long-term sensor baseline drift due to biofouling and thermal cycling
722 is corrected at three hardware and algorithmic levels. The primary fouling
723 barrier is the zwitterionic polymer functionalization layer deposited on the
724 graphene monolayer spacer of each NPoM cavity [42, 43], which physically
725 repels microbial adhesion and is modeled to preserve $\geq 90\%$ quantum coher-
726 ence after 60 d field immersion. As a secondary hardware barrier, a thin-film
727 micro-shielding layer (1 μm to 5 μm of SiO_2 or Al_2O_3) is deposited around the
728 NPoM picocavity and CQD sensor tip to insulate the quantum probe from
729 diurnal temperature swings. The shielding efficacy $\eta_{\text{shield}} = 0.85$ means the
730 probe experiences only 15% of the ambient greenhouse temperature fluctu-
731 ation, reducing the thermal contribution to Stern-Volmer drift from 5.25%
732 to 0.79% under a $\pm 15\text{K}$ diurnal cycle. The *DynamicCalibrator* acts as a
733 tertiary algorithmic correction: an exponential moving average (EMA) with
734 smoothing factor $\alpha_{\text{EMA}} = 0.1$ computes a multiplicative correction factor
735 $\kappa(t) = I_0^{\text{ref}}/\hat{\mu}(t)$ and raises a drift alarm when the fractional change in the
736 EMA $\hat{\mu}(t)$ exceeds 20% of the initial baseline, triggering OPV panel cleaning.

737 At the panel level, OPV electrical output is attenuated according to
738 Eq. (8), with a default decay rate of $0.5\% \text{d}^{-1}$ and a minimum soiling factor
739 of 0.75 (enforced floor at 25% maximum efficiency loss).

740 *Data availability*

741 All simulation parameters, YAML configuration files, orchestration
742 scripts, and analysis code needed to reproduce every figure and table are
743 available from the corresponding author upon reasonable request and will
744 be deposited in a public repository upon acceptance. The PT-HOPS/SBD
745 implementation is based on the open-source MesoHOPS library (v1.7) [20].

746 *Code availability*

747 The digital-twin simulation framework (orchestrator, QAOA scheduler,
748 LCA module, BB84 QKD simulator, SERS diagnostics, NV-diamond
749 pathogen sensor, and GQAS compliance checker) and all simulation pa-
750 rameters, including the 12-mode vibronic bath spectral density and the
751 8-site FMO electronic Hamiltonian specified in the machine-readable
752 `parameters.yaml` configuration file (v1.4), are available from the corre-
753 sponding author upon reasonable request. No public code or data archive
754 DOI has been minted yet.

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760 **CRedit authorship contribution statement**

761 **S.C. Teguia Kouam:** Conceptualization, Methodology, Software, For-
762 mal analysis, Writing – original draft. **T. Goumai Vedekoi:** Validation,
763 Writing – review & editing. **J.-P. Tchapel Njafa:** Validation, Writing –
764 review & editing. **J.-P. Nguenang:** Validation, Writing – review & editing.
765 **S.G. Nana Engo:** Supervision, Project administration, Writing – review &
766 editing.

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768 The authors declare that they have no known competing financial inter-
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780 content of the publication.

781 Nomenclature

Symbol	Meaning
Φ_{FT}	Forward (reaction-center) transfer yield
τ_c	Exciton coherence lifetime ($1/e$ decay of l_1 -norm coherence)
λ_{res}	Residual reorganization energy of polaron-dressed states
g_0 / g_{eff}	Single-emitter / collective plasmon–exciton coupling strength
κ / Q	Plasmon linewidth / quality factor
$\omega_{\text{pl}} / V_{\text{mode}}$	Plasmon frequency / NPoM mode volume
Γ_{RC}	Reaction-center trapping rate
$\text{EF}_{\text{SERS}} / g_{\text{om}}$	SERS enhancement factor / optomechanical coupling
$\text{ET}_c / K_c / f_{\text{shade}}$	Crop evapotranspiration / crop coefficient / OPV shading factor
NEB	Net Ecological Benefit
$\eta_{\text{soil}} / \eta_{\text{shield}}$	Soiling factor / micro-shielding efficacy
QBER	Quantum bit error rate (BB84 link)
Δg	Groundwater gravity anomaly
$\alpha / \alpha_{\text{EMA}}$	Sentinel area fraction / calibrator smoothing factor

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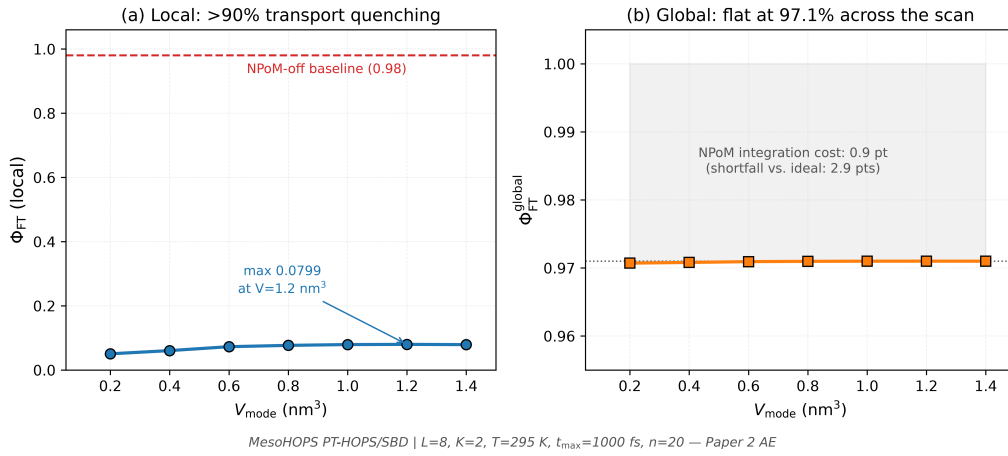


Figure 1: Local quenching, global stability: the central result in one picture. (a) Forward transfer yield Φ_{FT} across the NPoM mode-volume scan ($n = 20$, $T = 295$ K): locally, transport is quenched by more than 90 % relative to the pristine FMO baseline (0.98), with a factor-1.6 variation across the accessible range and a maximum of 0.0799 at $V_{mode} = 1.2$ nm³ (Table 2). (b) Area-weighted global canopy trapping yield for the 1 % sentinel fraction (Equation (5)): flat at 0.971 across the entire scan — the deep local suppression is absorbed at a constant 0.9-point canopy-scale cost (0.980 to 0.971), leaving the canopy 2.9 points below an ideal yield of 1.0. The flat curve in (b), not the quenching in (a), is the design-relevant outcome: diagnostics integrate into the stack without compromising productivity.

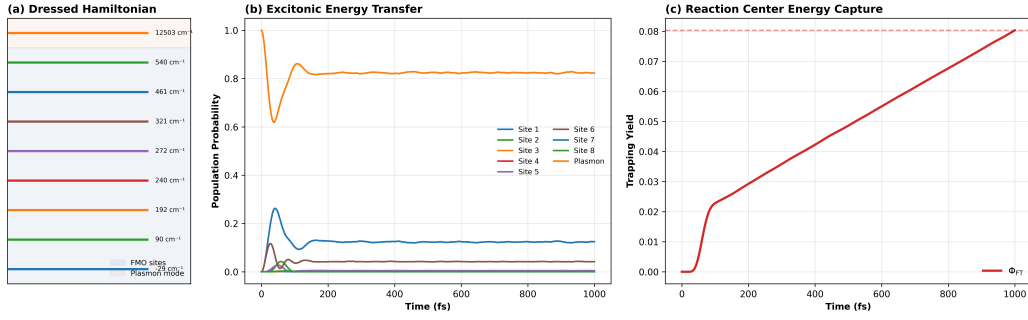


Figure 2: **Quantum dynamics of the 8-site FMO complex coupled to an NPoM plasmonic cavity within the agrivoltaic digital twin.** (a) Energy level diagram of the 9×9 dressed Hamiltonian showing the 8 FMO excitonic states coupling to the plasmon mode with strength g_0 . (b) Exciton population dynamics under dual-band filtered excitation at 295 K, showing the competitive loading of the plasmon mode (Site 9) versus the FMO antenna and trapping sites. (c) Reaction center trapping yield Φ_{FT} as a function of time. All data: $L = 8$, $K = 2$, $n = 100$ disorder realizations. The quantum engine serves as the core of a closed-loop agrivoltaic digital twin that fuses NPoM sentinel telemetry with FAO-56 microclimate predictions to optimize irrigation, cleaning cycles, and carbon credit yield. Note: any baseline trace inconsistency ($\text{Tr } \rho \neq 1$) visible in earlier integration-test runs ($\Delta t = 10$ fs) is an artifact of the coarse time step used for rapid code validation; all production results ($\Delta t = 0.2$ fs) preserve trace to 1.000 000 (Section S11.2).

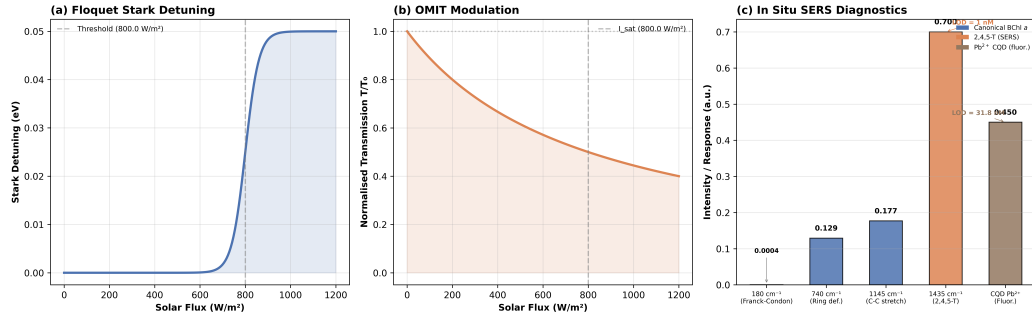


Figure 3: Dynamic photo-protection, in situ diagnostics, and environmental robustness. (a) Floquet Stark detuning amplitude as a function of solar flux. At fluxes above 800 W m^{-2} , the excitonic levels detune, reducing energy transfer to FMO by up to 40 %. (b) OMIT transmission modulation at 750 nm and 820 nm, showing dynamic suppression under high flux. (c) Simulated SERS Raman spectrum showing the canonical BChl *a* vibrational modes (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}). The three agriculturally specific target signatures assigned to the sentinel sub-system — oxidative stress (1145 cm^{-1}), 2,4,5-T pesticide contamination (1435 cm^{-1} , $\text{LOD} = 1 \text{ nmol L}^{-1}$), and Pb^{2+} via encapsulated CdTe/ZnSe quantum dot fluorescence quenching ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$) — are detailed in the main text and SI. A DynamicCalibrator using exponential moving average ($\alpha = 0.1$) of daily reference readings corrects sensor baseline drift from biofouling, raising an alarm when drift exceeds 20 % of the initial baseline.

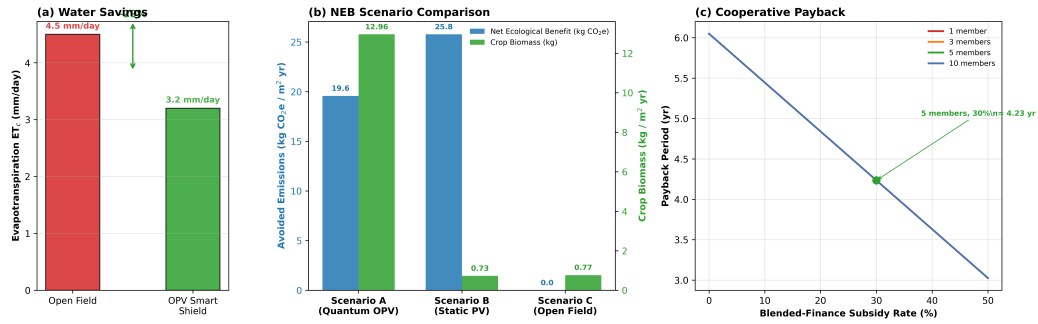


Figure 4: **Water–Energy–Food network assessment.** (a) FAO-56 evapotranspiration comparison: open field ($ET_c = 4.5 \text{ mm d}^{-1}$) vs. OPV smart shield ($ET_c = 3.2 \text{ mm d}^{-1}$), a 28 % reduction. (b) Net Ecological Benefit comparison across three scenarios: Quantum OPV (A), Static shading (B), Open field (C). Scenario A achieves $19.6 \text{ kg CO}_2\text{e m}^{-2} \text{ yr}^{-1}$ with a crop biomass of $12.0 \text{ kg m}^{-2} \text{ yr}^{-1}$, while Scenario B reaches $25.8 \text{ kg CO}_2\text{e m}^{-2} \text{ yr}^{-1}$ but only $0.68 \text{ kg m}^{-2} \text{ yr}^{-1}$ in biomass. The quantum shield preserves near-baseline crop productivity (0.971 global trapping yield) whereas static shading suppresses it to 0.064, quantifying the yield–carbon trade-off. The NEB serves as the verifiable metric for a tiered subsidy framework that bridges the quantum divide. (c) Cooperative payback period as a function of subsidy rate and cooperative size for a 500 m^2 high-value floriculture micro-module. For 5 members with 30 % blended-finance subsidy, payback = 4.23 yr (net annual cashflow of $26\,612 \text{ USD yr}^{-1}$ after OPEX). This indicates that, under the modelled cooperative assumptions, quantum-guided agrivoltaics can shorten payback when coupled with cooperative ownership and strategic crop selection.

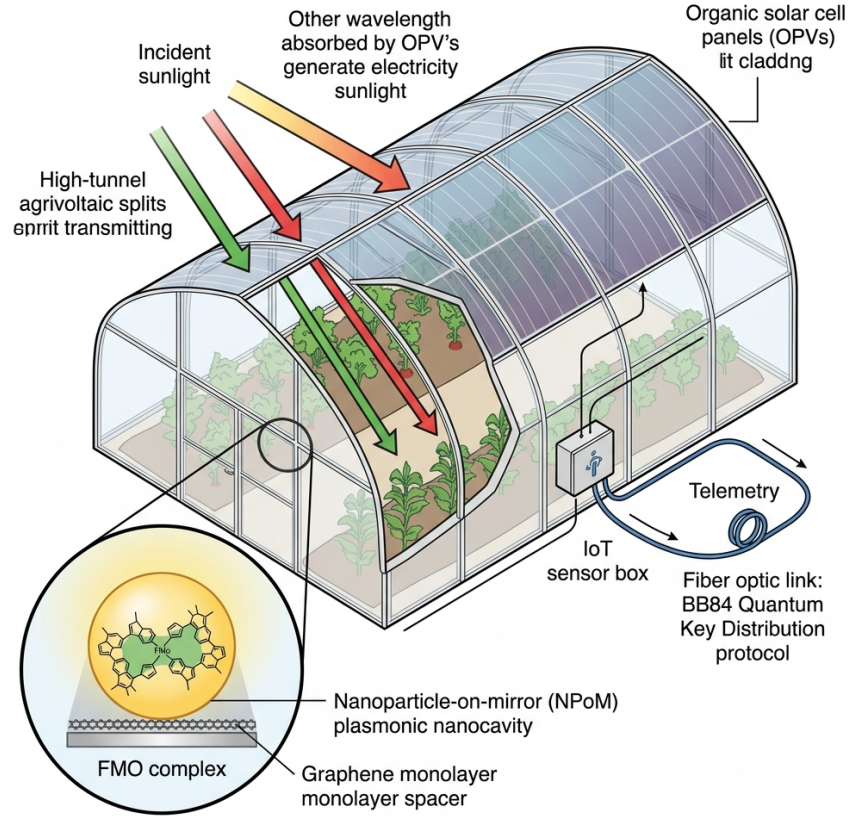


Figure 5: **Schematic layout of the multi-scale quantum agrivoltaic digital twin platform.** Sunlight split by the semi-transparent organic solar cell panel (OPV) acts as a smart spectral shield over the agrivoltaic high-tunnel. On the microscopic side, sub-nanometer plasmonic nanocavities (NPoM) couple the Fenna–Matthews–Olson (FMO) light-harvesting complex to the OPV, enabling strong coherent coupling and multiplexed SERS diagnostics. Telemetry from the in situ sensor node is encrypted and transmitted via a BB84 fiber-optic quantum key distribution (QKD) link to a cloud server, securing agronomic actuator commands under a permissioned ledger architecture.

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